QUANT

Number System

(Digital Root, Unit's Digit, Last two digits
& other important concepts)

Assignment 02

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Numbers Online Assignment -02

1. By which one of the following should 152207 be multiplied so that the product is 11111111?

1. 53 2. 63 3. 73 4. 87

2. What is the sum of the odd natural numbers up to 70?

1. 1120 2. 1190 3. 1225 4. 1260

3. After the division of a number successively by 3, 4 and 7, the remainders obtained are 2, 1 and 4 respectively. What will be the remainder if 84 divides the same number?

1. 80 2. 53 3. 41 4. 76

4. Q is a prime number, then Q can have how many values if $(Q^4 + 7)$ is also a prime number?

1. 3 values 2. 4 values 3. 2 values 4. 1 value

5. If a person selects a non-composite number out of the first 100 natural numbers in such a way that twice of that number is a prime number, what can you conclude about such a number?

1. the number is prime
2. the number is more than 5
3. the number is less than 5
4. the number is a two-digit number

6. Which of the following cannot be the sum of three consecutive prime numbers?

1. 269 2. 210 3. 301 4. 131

7. $J = 172 \times 189 \times 279 \times 871 \times 693$. What will be the remainder when J is divided by 11?

1. 2 2. 0 3. 10 4. 9

8. A certain no. is divided by 552 & 28 was obtained as remainder, what would be the remainder if same no. is divided by 23?

1. 3 2. 5 3. 21 4. Cant Determined

9. In a party there are 30 persons present, & everybody shakes hand with all the other persons. What is the total no. of handshakes happening?

1. 420 2. 425 3. 435 4. 465

10. Find the last digit of the number $N = 1^3 + 2^3 + 3^3 \ldots\ldots\ldots\ldots\ldots + 99^3$.

1. 0 2. 1 3. 2 4. 5

Answer keys & Explanations

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1.	3	One method in such questions is to check the unit's digit, which deletes the 4th option. Then if approximate values are taken, it can be seen that 53 × 11 cannot provide 100 +. Only 73, when multiplied can give something 100+. Hence you are left with the 3rd option.
2.	3	Here, one way of thinking is making an AP. But there is a straight way. Make the students note the points. The sum of first n odd natural numbers is n^2. Whereas the sum of first n even natural numbers is $n^2 + n$.
3.	2	Before the third division the number was 7x + 4. Before the second division the number was (7x + 4) × 4 + 1. Before the first division the number was [(7x + 4) × 4 + 1] × 3 + 2. Simplifying this expression, we get 84x + 53. If we divide this number by 84, the first term will yield a zero remainder, and the remainder will be 53.
4.	4	Go to the last part of the question. (Q^4 + 7) cannot be even, as there is only 1 even prime number. When it is odd, this implies Q^4 must be even and it can be even only if Q itself is even. Now Q is a prime number also. So the only possible value Q can take is 2. Thus 4th option is the answer.
5.	3	When twice of a number is prime, this implies the prime number involved must be even. There is only one even prime, which is twice of 1 ⇒ 1 is the non-composite number he is referring to and 3rd option is answer.
6.	2	When the three consecutive prime numbers are used, they must give an odd total if 2 is not one of them. As the totals given are more than 200, this means all the prime number used are in the range of 60 – 80. They must give an odd total only and thus 2nd option is the answer.
7.	2	Instead of multiplying & then dividing, it is better to divide each number and get the remainders. If carefully observed, it can be seen that the last number 693 is divisible by 11, therefore J will be completely divisible by 11 and will give 0 as the remainder.
8.	2	The no. is of the form of 552Q + 28 and it is to be divided by 23. As 552 is divisible by 23, thus whatever is the value of Q, it will not provide any divisor. Now divide 28 by 23 and get the remainder as 5, which is your answer **Note:** In case the second divisor is not a factor of the 1st divisor, then you cannot solve this question.
9.	3	The 1st person will shake hand with 29 persons, the 2nd with 28 and so on. Thus sum of 1st 29 natural numbers is to be calculated. Hence 435 is the answer.
10.	1	$1^3 + 2^3 + 3^3$ ……. + 99^3 is the addition of cubes of 1st 99 natural numbers. Using the formula of $\sum N^3$, we get the answer as $[(99 \times 100)/2]^2$ which would give the last digit as zero.